

# 2023 Ch H2 Q3

Section: Nature's Chemistry

Topic: Proteins and Fats

## Question Summary

Cheese is a complex mixture of fats, proteins, and flavour molecules. (a) Questions on fats and fatty acids: glycerol, condensation reactions, functional groups, oxidation, hydroxycarboxylic acids, and gas chromatography. (b) Proteins: amino acids in casein, essential amino acids, denaturation. (c) Role of emulsifiers in processed cheese. (d) Naming of calcium propanoate as a preservative.

## Worked Solution

According to the SQA Marking Instructions for 2023 H2 Q3:

(a)(i) (A) Glycerol = systematic name propane-1,2,3-triol. (B) Fats are formed by condensation (esterification) between fatty acids and glycerol.

(a)(ii) (A) Circled group in X is a hydroxyl group. (B) Molecule Y is an alkene formed by dehydration. (C)  $Y \rightarrow Z$  involves hydrogenation (addition of  $H_2$ ). (D) Reaction 3 increases the oxygen-to-hydrogen ratio (oxidation).

(a)(iii) Hydroxycarboxylic acids contain  $-OH$  and  $-COOH$  groups in the same molecule. The acid that forms the given cyclic ester is 4-hydroxybutanoic acid.

(a)(iv) (A) Retention time missing from Chromatogram Y is 30 minutes. (B) To quantify flavour molecule 3, the sample must be diluted so the peak lies within the calibration range.

(b)(i) Amino acids such as leucine and valine that must be obtained from the diet are essential amino acids. (b)(ii) Glutamic acid structure:

$HOOC-CH_2-CH_2-CH(NH_2)-COOH$ . (b)(iii)

Heating proteins causes them to denature (loss of 3D shape).

(c) Function of an emulsifier: allows mixing of oil and water phases by reducing surface tension and forming micelles.

(d)  $\text{Ca}^{2+}(\text{CH}_3\text{CH}_2\text{COO}^-)_2$  is calcium propanoate.

### **Final Answer**

**(a)(i) Propane-1,2,3-triol; condensation (esterification).**

**(a)(ii) Hydroxyl group; alkene; hydrogenation; reaction 3 oxidation.**

**(a)(iii) 4-hydroxybutanoic acid.**

**(a)(iv) 30 minutes; dilute the sample.**

**(b)(i) Essential amino acids.**

**(b)(ii) Glutamic acid structure  $\text{HOOC}-\text{CH}_2-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{COOH}$ .**

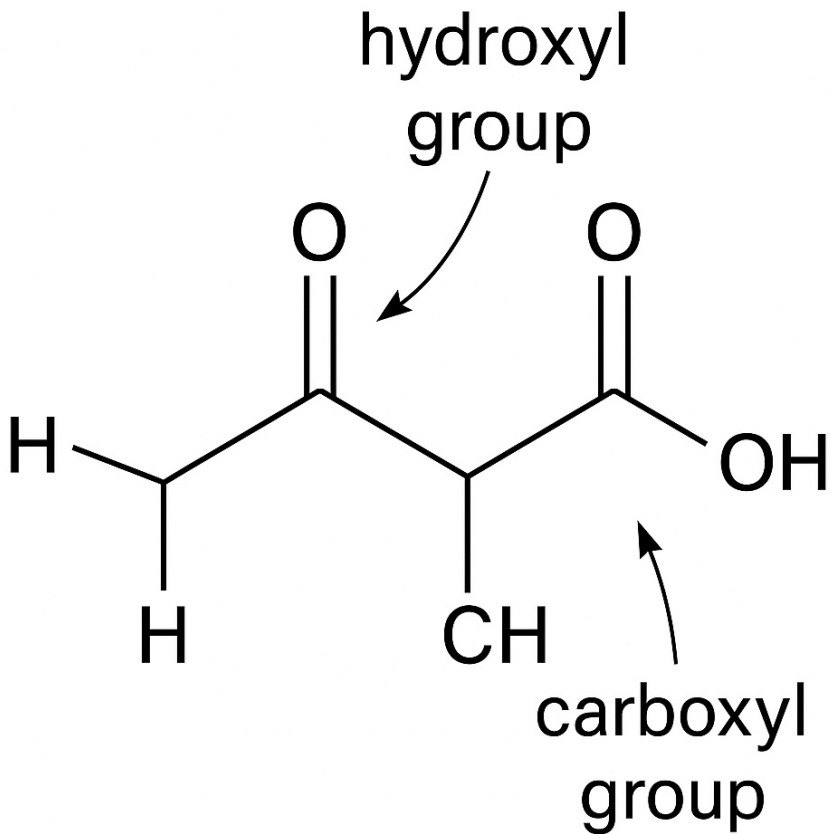
**(b)(iii) Denaturation.**

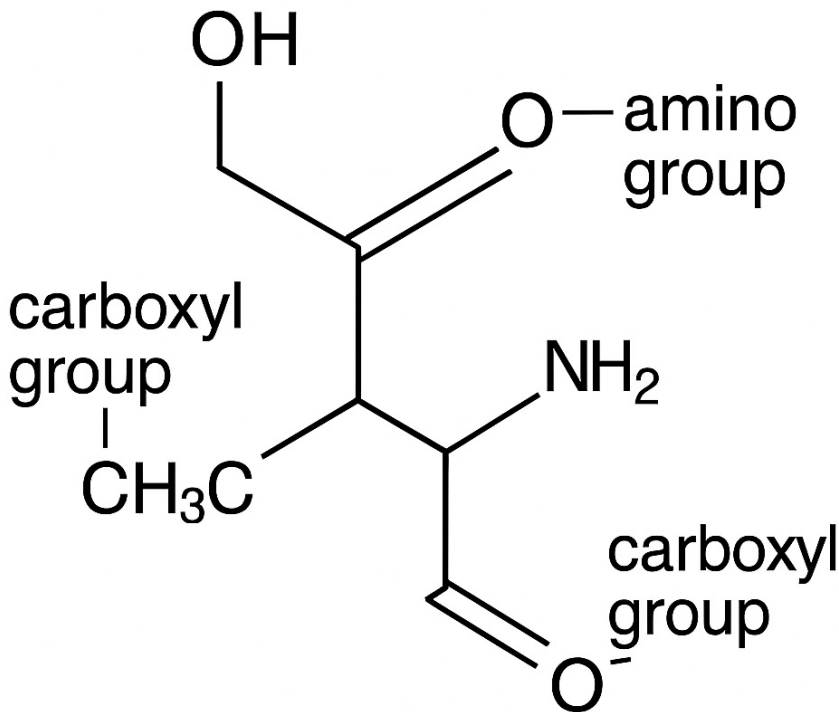
**(c) Emulsifier allows mixing of oil and water phases.**

**(d) Calcium propanoate.**

### **Revision Tips**

- Glycerol = propane-1,2,3-triol; fats form by condensation with fatty acids. • Hydroxycarboxylic acids can cyclise to esters. • Gas chromatography: retention time and peak area give identity and concentration. • Essential amino acids cannot be synthesised by the body. • Emulsifiers stabilise oil-water mixtures. • Calcium propanoate is a common food preservative.





glutamic acid